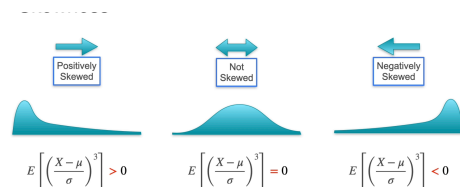


Skewness and Kurtosis

Skewness

the skewness tell us whether a distribution leans to one side.

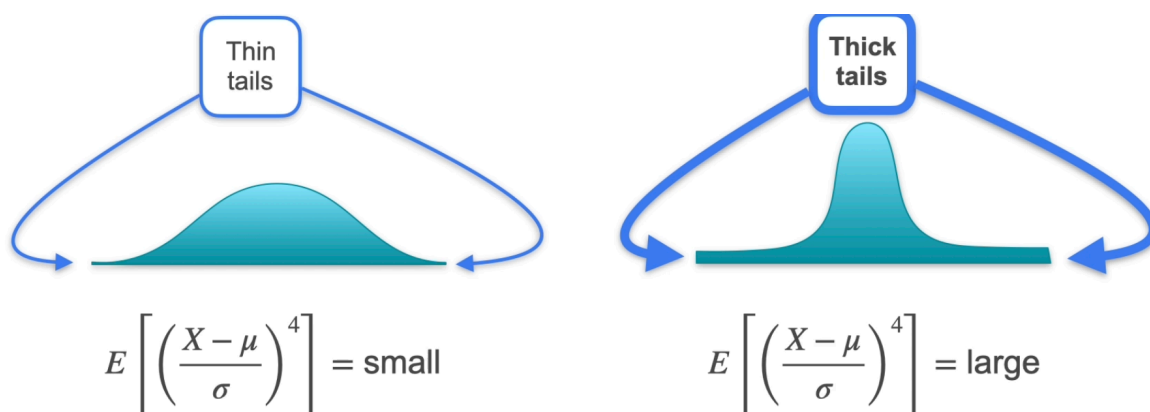


The mean means nothing or misleading if the data is skewed. and the median is often a better measurement for the skewed data

Kurtosis

Kurtosis is the fourth moment

It tell us how much of the data lives in the tails vs the center



Even if they have the same variance!

The fat tail mean extreme events are more likely than those models predict, this is crucial in finance, insurance, and the risk management.